Determinants of immune responses predictive of protection against shigellosis in an endemic zone: a systems analysis of antibody profiles and function

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Summary

Background ***Shigella*** **is the third leading global cause of moderate or severe diarrhoea among children younger than 5 years globally, and is the leading cause in children aged 24–59 months. The mechanism of protection against** ***Shigella*** **infection and disease in endemic areas is uncertain. We aimed to compare the** ***Shigella*****-specific antibody responses in individuals living in** ***Shigella*****-endemic and non-endemic areas, and to identify correlates of protection in a** ***Shigella*****-endemic location.**

Methods We applied a systems approach to retrospectively analyse serological responses to ***Shigella*** **across endemic and non-endemic populations. We profiled serum samples collected from 44 individuals from the USA without previous exposure to** ***Shigella*** **and who were experimentally challenged with** ***Shigella sonnei*** **(non-endemic setting), and serum samples collected from 55 Peruvian army recruits (endemic setting). In the endemic setting, a subset of 37 samples collected from individuals infected with culture-confirmed** ***Shigella flexneri*** **2a were divided into two groups: susceptible, which included individuals infected within 90 days of entering the camp (n=29); or resistant, which included individuals infected later than 90 days after entering the camp (n=8). We analysed** ***Shigella*****-specific antibody isotype, subclass, and Fc receptor binding profiles across IpaB, IpaC, IpaD, and lipopolysaccharide from** ***S flexneri*** **2a, 3a, and 6, and** ***S sonnei*****, and O-specific polysaccharide (OSP) from** ***S flexneri*** **2a and 3a and** ***S sonnei*****. We also evaluated antibody-mediated complement deposition and innate immune cell activation. The main outcome of interest was the detection of antibody markers and functionality associated with protection against shigellosis in a high-burden endemic setting.**

Findings Adults with endemic exposure to ***Shigella*** **possessed broad and functional antibody responses across polysaccharide, glycolipid, and protein antigens compared with individuals from non-endemic regions. In a setting with high** ***Shigella*** **burden, elevated levels of OSP-specific Fcα receptor (FcαR) binding antibodies were associated with resistance to shigellosis, whereas total OSP-specific IgA was not, suggesting a potentially unique functionality. OSP-specific FcαR binding IgA found in resistant individuals activated bactericidal neutrophil functions including phagocytosis, degranulation, and production of reactive oxygen species. Moreover, IgA depletion from resistant serum significantly reduced binding of OSP-specific antibodies to FcαR and antibody-mediated activation of neutrophils and monocytes.**

Interpretation Our findings suggest that OSP-specific functional IgA responses contribute to protective immunity against ***Shigella*** **infection in a high-burden setting. These findings will assist in the development and evaluation of** ***Shigella*** **vaccines.**

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Introduction

Diarrhoeal diseases caused by enteric pathogens are a major cause of death in children younger than 5 years in low-income and middle-income countries.[1,2] Attempts to develop effective vaccines against enteric pathogens have been impeded by both the breadth of pathogens involved, and incomplete understanding of both correlates and mediators of protective immunity against individual pathogens in endemic settings. Vaccines that have been protective against enteric pathogens, including rotavirus,[3] *Shigella*,[4] and *Vibrio cholerae*,[5] in non-endemic settings have provided decreased protection when evaluated in endemic areas. Individuals living in *Shigella*-endemic regions are exposed to *Shigella* infection from childhood and develop partially protective immunity against shigellosis during the first years of life.[6] However, our understanding of the immune response that evolves in these individuals is incomplete. Deciphering correlates of protection against

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Evaluation of 3β-hydroxysteroid dehydrogenase activity using progesterone and androgen receptors-mediated transactivation

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3β-Hydroxysteroid dehydrogenases (3β-HSDs) catalyze the oxidative conversion of delta (5)-ene-3-beta-hydroxy steroids and ketosteroids. Human 3β-HSD type 2 (HSD3B2) is predominantly expressed in gonadal and adrenal steroidogenic cells for producing all classes of active steroid hormones. Mutations in HSD3B2 gene cause a rare form of congenital adrenal hyperplasia with varying degree of salt wasting and incomplete masculinization, resulting from reduced production of corticoids and androgens. Therefore, evaluation of the HSD3B2 enzymatic activity in both pathways for each steroid hormone production is important for accurately understanding and diagnosing this disorder. Using progesterone receptor (PR)- and androgen receptor (AR)-mediated transactivation, we adapted a method that easily evaluates enzymatic activity of HSD3B2 by quantifying the conversion from substrates [pregnenolone (P5) and dehydroepiandrosterone (DHEA)] to (progesterone and androstenedione). HEK293 cells were transduced to express human HSD3B2, and incubated medium containing P5 or DHEA. Depending on the incubation time with HSD3B2-expressing cells, the culture media progressively increased luciferase activities in CV-1 cells, transfected with the PR/AR expression vector and progesterone-/androgen-responsive reporter. Culture media from human and